

一. ① $9s^2 + 20s + 1$ 中 a_0, a_1, a_2 均大于 0, $\therefore$ 稳定

$$\begin{array}{rcl} s^3 & 1 & 11 \\ s^2 & 6 & K \\ s^1 & \frac{66-K}{6} & \\ s^0 & K & \end{array} \quad \begin{cases} \frac{66-K}{6} > 0 \\ K > 0 \end{cases} \Rightarrow \therefore 0 < K < 66$$

令 $z = s + 0.5$, 则 $s = z - 0.5$

代入原多项式, 得: $(z-0.5)^3 + 6(z-0.5)^2 + 11(z-0.5) + K = z^3 + 4.5z^2 + 5.75z - 4.125 + K$

$$\begin{array}{rcl} z^3 & 1 & 5.75 \\ z^2 & 4.5 & K-4.125 \\ z^1 & \frac{30-K}{4.5} & \\ z^0 & K-4.125 & \end{array} \quad \begin{cases} \frac{30-K}{4.5} > 0 \\ K-4.125 > 0 \end{cases} \Rightarrow 4.125 < K < 30$$

② 二阶系统单位斜坡响应的稳态误差为 $\frac{2\xi}{\omega_n}$

$$r(t) = 4t \Rightarrow R(s) = \frac{4}{s^2} \quad \omega_n^2 = 10 \Rightarrow \omega_n = \sqrt{10} \text{ rad/s} \quad \xi = \frac{4}{2 \times \sqrt{10}} = \frac{2}{\sqrt{10}}$$

$$\therefore e_{ss} = 4 \times \frac{2\xi}{\omega_n} = \frac{8 \times \frac{2}{\sqrt{10}}}{\sqrt{10}} = 1.6$$

二. ① $p_1 = 0, p_2 = -1, p_3 = -2$

$$\text{渐近线: } \sigma_A = \frac{p_1 + p_2 + p_3}{3} = \frac{-1-2}{3} = -1$$

$$-180^\circ, 60^\circ, -60^\circ$$

分离点:

$$\frac{1}{6B} + \frac{1}{6B+1} + \frac{1}{6B+2} = 0$$

$$6B = \frac{-6 \pm 2\sqrt{3}}{3 \times 2} = -1 \pm \frac{\sqrt{3}}{3}$$

$\therefore 6B$ 必须在 -1 和 0 之间, $\therefore 6B \approx -0.423$

分离角: $\pm 90^\circ$

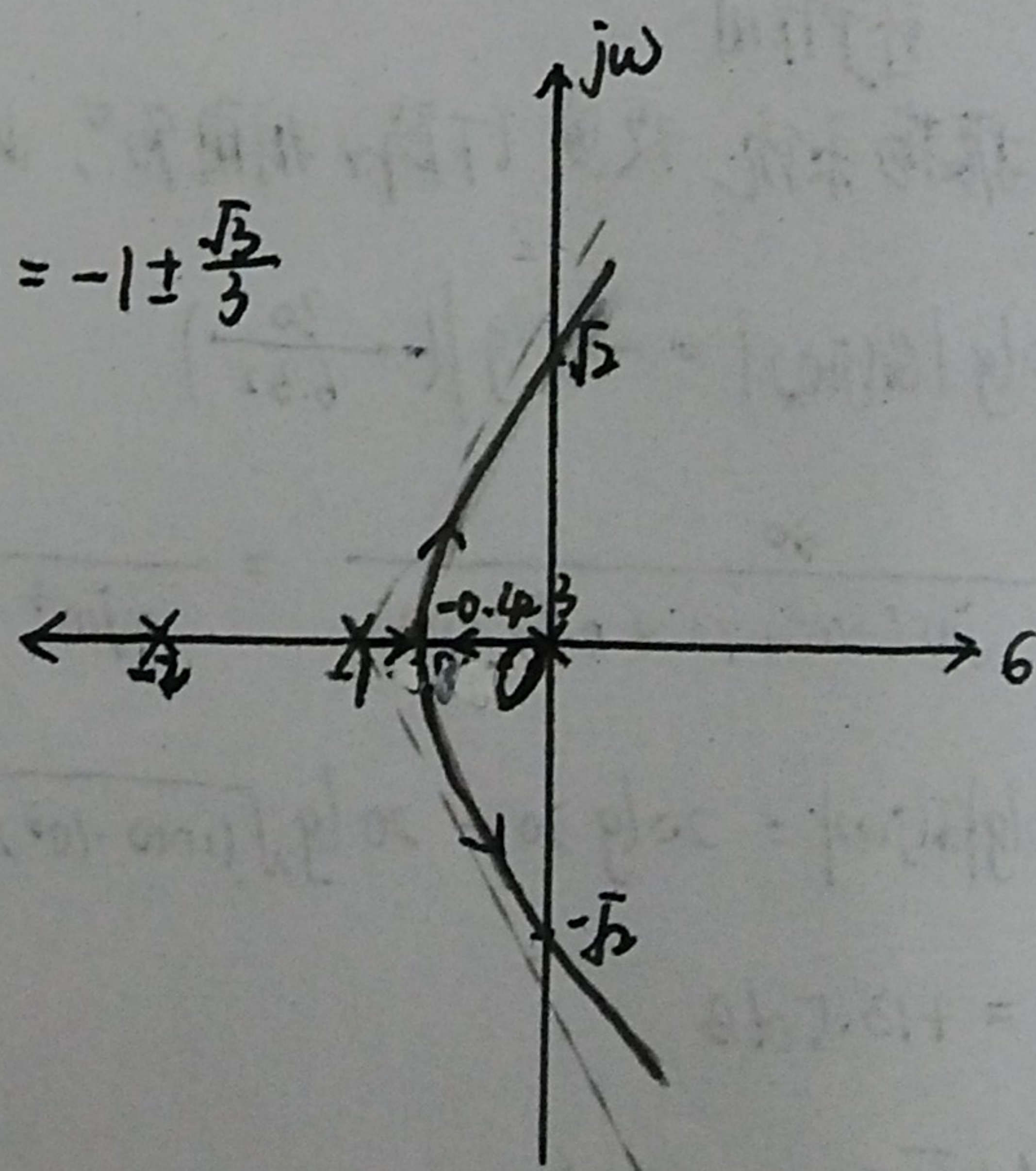
$$D(s) = s(s+1)(0.5s+1) + K = 0 \quad \text{令 } s = j\omega$$

$$\text{可得: } -1.5\omega^2 + K + j(\omega - 0.5\omega^3) = 0$$

$$\begin{cases} K - 1.5\omega^2 = 0 \\ \omega - 0.5\omega^3 = 0 \end{cases} \Rightarrow \omega_1 = 0 \text{ 或 } \omega_2 = -\sqrt{2} \text{ 或 } \omega_3 = \sqrt{2} \quad (\text{舍去})$$

② 由 $K = |s+1||0.5s+1|$ 将 $s_{1,2} = -0.33 \pm j0.57$ 代入式中得:

$$K = |-0.33 \pm 0.57j| \cdot |0.67 \pm 0.57j| \cdot |1 + 0.835 \pm 0.285j| = 0.659 \times 0.880 \times 0.882 = 0.511$$



$$G(s) = \frac{2K}{s(s+1)(s+2)}$$

由特征根之积可知:

$$\prod_{i=1}^3 p_i = (-1)^3 \cdot 2K \Rightarrow (-0.33 + j0.57)(-0.33 - j0.57) \cdot p_3 = -2 \times 0.511 \Rightarrow p_3 = -2.35$$

∴ 可见, 主导极点为 $s_{1,2} = -0.33 \pm j0.57$

那么可简化系统为:

$$G(s) = \frac{K^*}{s(s+1)} \quad \left\{ \begin{array}{l} K^* = 0.659 \times 0.88 = 0.58 \\ \Rightarrow G(s) = \frac{0.58}{s(s+1)} \end{array} \right.$$

三. 题中缺条件, 不妨令转折点为 $\omega_n^2 = 10$

① 积分环节 $\frac{1}{s}$, 比例环节 K , 振荡环节 $\frac{1}{\frac{s^2}{\omega_n^2} + 2\xi\frac{s}{\omega_n} + 1}$

$$\text{分析: } \omega_n = \sqrt{10} = 3.16 \text{ rad/s} \quad -20\lg 2\xi = 14 \text{ dB} \Rightarrow \xi = 0.1$$

$$-20\lg \omega_0 + 20\lg K = 0 \Rightarrow K = 3$$

$$\therefore G(s) = \frac{3 \times (3.16)^2}{s(s^2 + 2 \times 3.16 \times 0.1s + 3.16^2)} = \frac{30}{s(s^2 + 0.632s + 10)}$$

② $N = -1$ 显然 $G(s)$ 在右半平面的极点个数加 0, ∴ 不稳定

③ 增益裕度:

由分析可知:

环节作用

在 ω_n 时, 振荡系统改变 (下降) 相角 90° , 此时 $\varphi(\omega) = -180^\circ$

$$K_g = -20\lg |G(j\omega)|$$

$$G(j\omega) = \frac{30}{j\omega(-\omega^2 + 10 + 0.632j\omega)} = \frac{30}{-j\omega^3 + 10j\omega - 0.632\omega^2} = \frac{30}{j(10\omega - \omega^3) - 0.632\omega^2}$$

$$A(\omega) = 20\lg |G(j\omega)| = 20\lg 30 - 20\lg \sqrt{(10\omega - \omega^3)^2 + (0.632\omega^2)^2}$$

$$\therefore A(\sqrt{10}) = +13.5 \text{ dB}$$

$$\therefore K_g = -A(\sqrt{10}) = -13.5 \text{ dB}$$

含义: 频率 ω_g (此时相角为 180°), 使系统 达到临界稳定所增大的幅值倍数

$$4. \textcircled{1} G(z) = Z \left[\frac{1-e^{-Ts}}{s} \cdot \frac{1}{s+2} \right] = (1-z^{-1}) Z \left[\frac{1}{s(s+2)} \right]$$

$$\because \frac{1}{s(s+2)} = \left(\frac{1}{s} - \frac{1}{s+2} \right) \times \frac{1}{2}$$

$$\therefore Z \left[\frac{1}{s(s+2)} \right] = \frac{1}{2} \left[\frac{1}{1-z^{-1}} - \frac{1}{1-e^{-2T}z^{-1}} \right] = \frac{z(1-e^{-2T})}{2(z-1)(z-e^{-2T})} = \frac{0.8647z}{2(z-1)(z-0.8647)} = \frac{0.43235}{(z-1)(z-0.8647)}$$

$$G(z) = \frac{z-1}{z} \times \frac{0.43235}{(z-1)(z-0.8647)} = \frac{0.43235}{z-0.8647}$$

$$\textcircled{2} T(z) = \frac{G(z)}{1+G(z)} = \frac{0.43235}{z-0.43235}$$

$$\textcircled{3} \overline{T}(z) = \frac{C(z)}{R(z)} = \frac{1}{1 + \frac{z}{z-e^{-2T}}} = \frac{z-e^{-2T}}{2z-e^{-2T}}$$

$$C(z) = R(z) \cdot T(z) = \frac{Tz}{(z-1)^2} \times \frac{z-e^{-2T}}{2z-e^{-2T}} = \frac{z(z-0.1353)}{(z-1)^2(2z-0.1353)}$$

五. ① 可控性

$$Q_c = [B \ AB] = \begin{bmatrix} 1 & -1 \\ 1 & -3 \end{bmatrix} \Rightarrow \text{rank } Q_c = 2 \therefore \text{可控}$$

② 可观性

$$Q_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ -\frac{3}{2} & -\frac{3}{2} \end{bmatrix} \Rightarrow \text{rank } Q_o = 2 \therefore \text{可观}$$

$$\textcircled{2} (sI-A) = \begin{bmatrix} s+1 & 0 \\ 0 & s+3 \end{bmatrix}, \det(sI-A) = (s+1)(s+3) = s^2+4s+3 \Rightarrow a_1=4, a_0=3$$

$$P^{-1} = Q_c L = \begin{bmatrix} 1 & -1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$$

$$C_s = CP^{-1} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \end{bmatrix}$$

配闭环极点, 在 $-3 \pm j$ 处, 即闭环特征方程为 $(s+3-j)(s+3+j)=0 \Rightarrow s^2+6s+10=0$

$$\therefore T(s) = \frac{s+2s}{s^2+6s+10}$$

$$\text{原有的特征方程为: } s^2+4s+3=0 \Rightarrow s^2+(4-h_1)s+(3-h_0)=s^2+6s+10$$

$$\therefore h_1=-2, h_0=-7 \Rightarrow \text{向量 } H = \begin{bmatrix} -7 & -2 \end{bmatrix}$$

反馈框图:

